

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November – December 2011

EC 202 Network Theory

Date: 30.11.2011, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

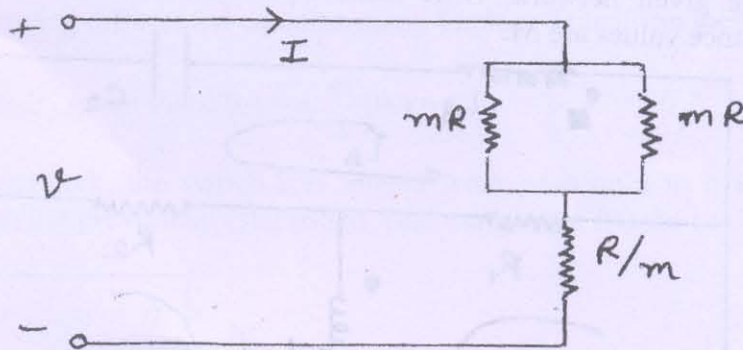
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

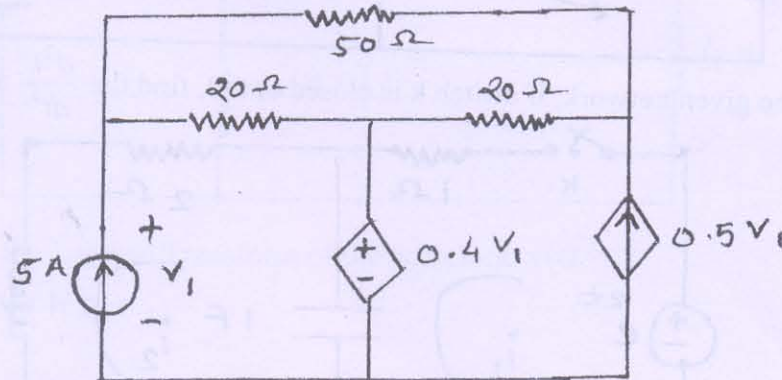
SECTION - I

Q - 1 (a) Find the value of m so that current I through the given network is maximum. (R is constant) [04]

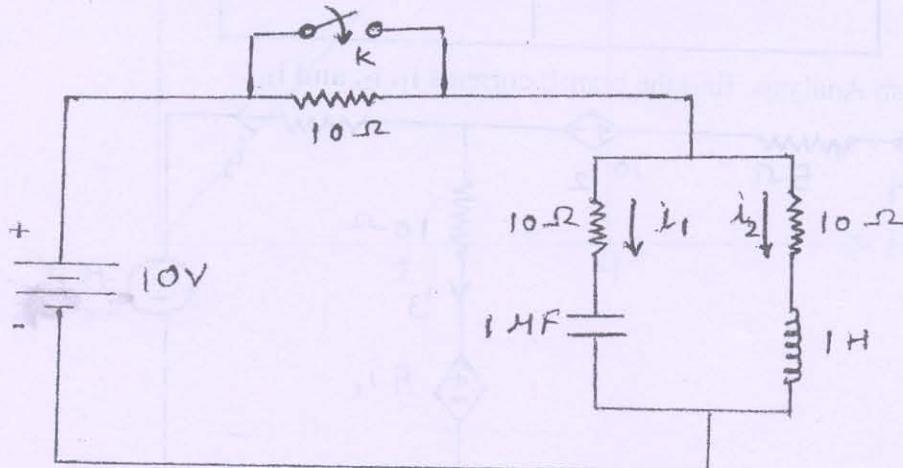


(b) Explain the Source Transformation rules (any five). [03]

Q - 2 (a) Using Nodal Analysis, find the voltage V_1 in the following Network. [04]

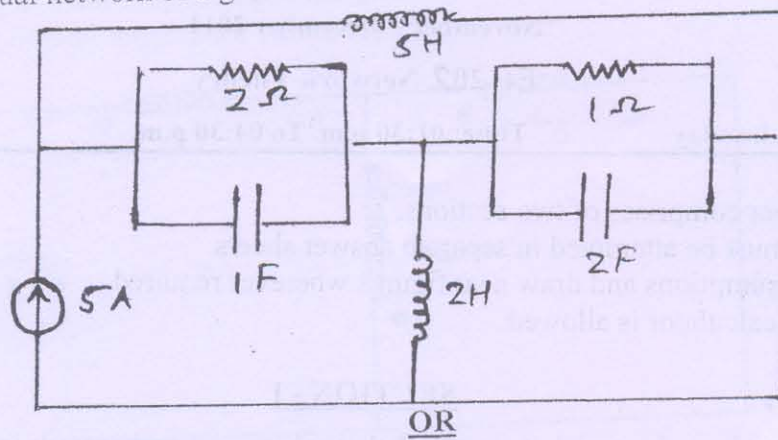


(b) In the given Network, steady state is reached with the switch k open. At time $t=0$, switch k is closed. find the value of i_1 , i_2 , di_1/dt , di_2/dt at $t=0+$. [07]



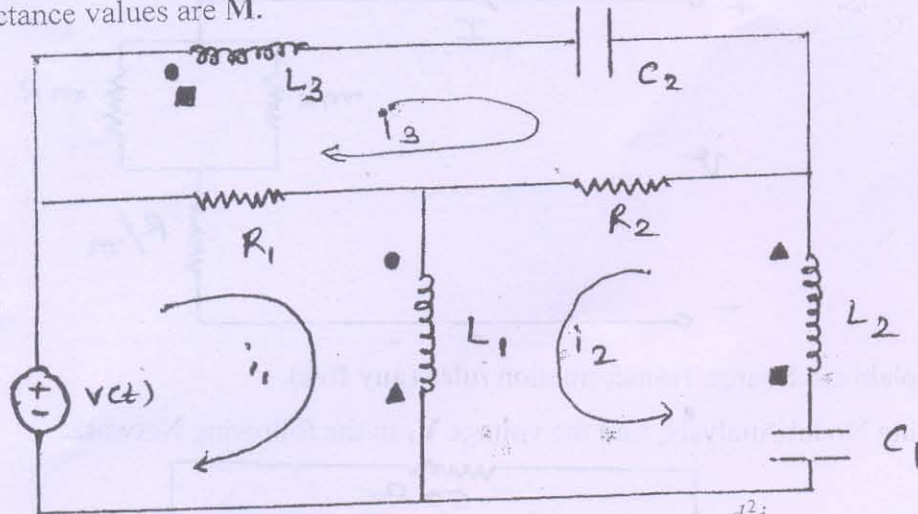
(c) Find the Dual network of a given network.

[03]

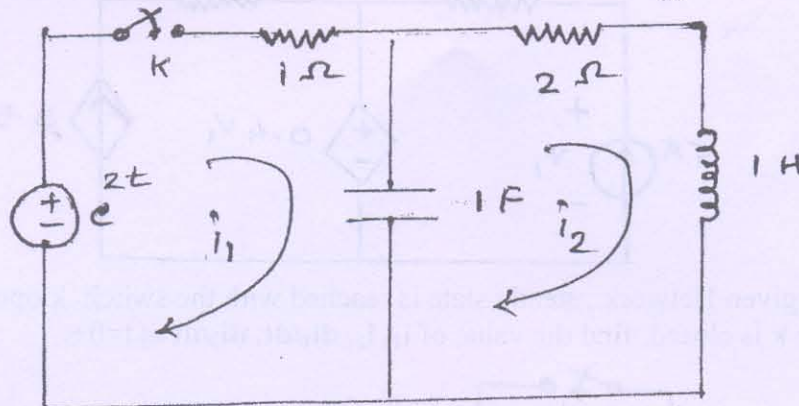


OR

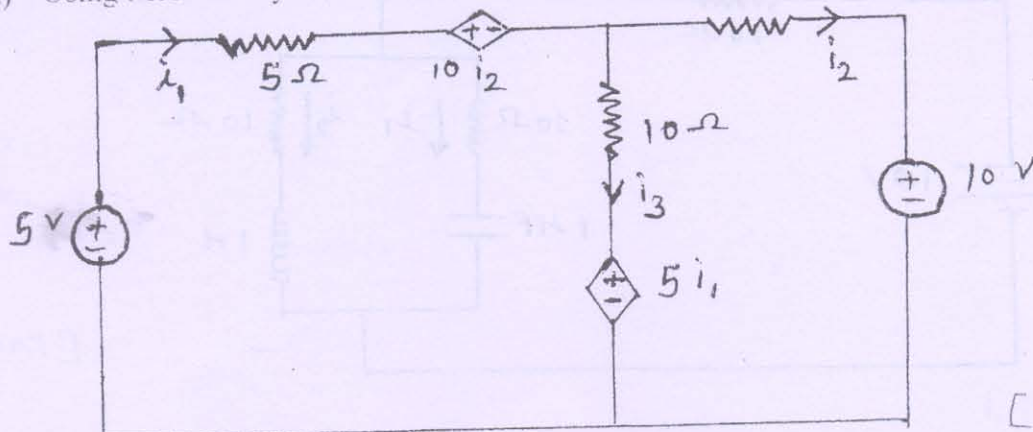
Q-2 (a) For the given network, write mesh equations for three loops, where all mutual inductance values are M. [07]



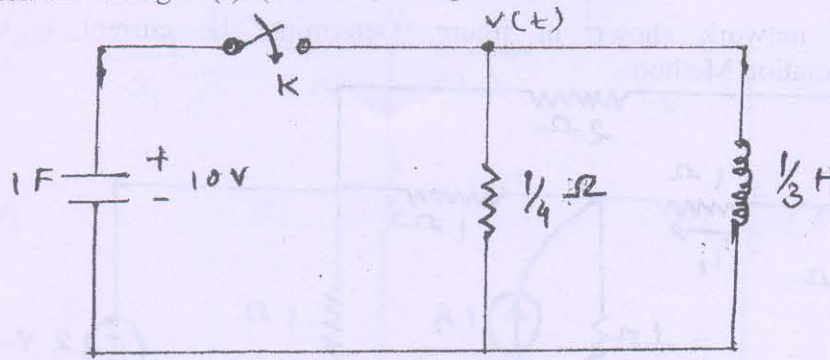
(b) For the given network, if switch k is closed at $t=0$, find the $\frac{d^2 i_1}{dt^2}$ at $t=0+$. [07]



Q-3 (a) Using Mesh Analysis, find the branch currents i_1 , i_2 , and i_3 . [07]



- (b) In the given network, capacitor C has an initial voltage $V_c(0_-) = 10 \text{ V}$ and at the same instant, current through the inductor is zero, the switch k is closed at $t=0$. Obtain the solution for voltage $v(t)$. (Do not use Laplace Transform). [07]

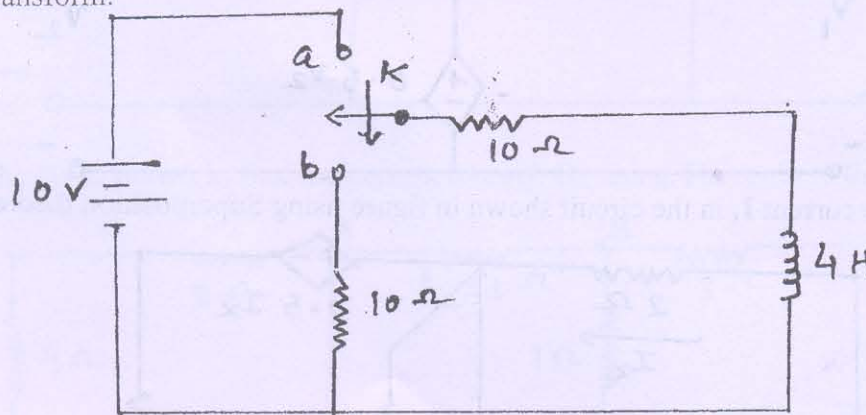


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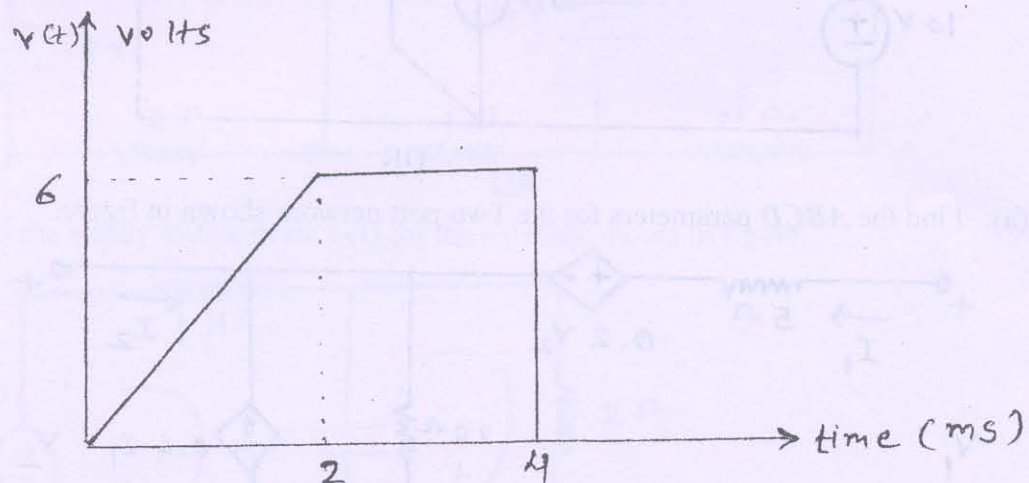
- Q - 3 (a) Solve the following differential equation using Laplace Transform method. [07]

$$\frac{d^2 i}{dt^2} + \frac{di}{dt} = t^2 + 2t \quad \text{given that } i(0_-) = 4 \quad \frac{di}{dt}(0_-) = -2.$$

- (b) In the given network, the switch k is moved from position a to b at time $t=0$, the steady state having previously established, find the current $i(t)$ for $t \geq 0$ using Laplace Transform. [04]

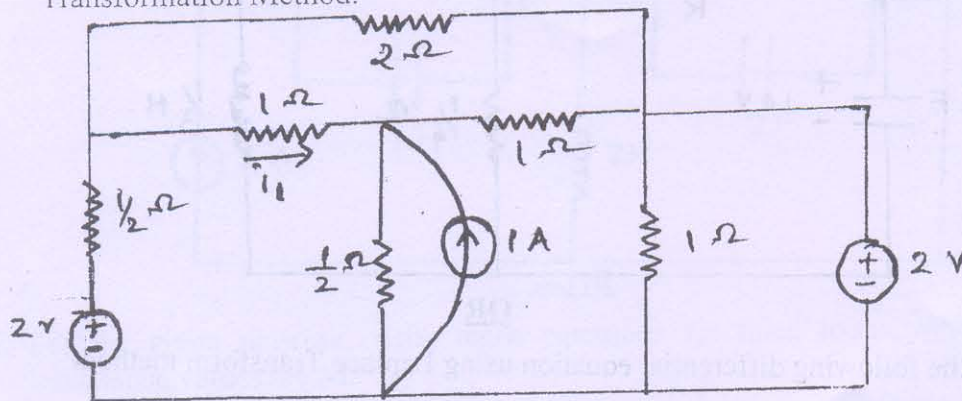


- (c) Determine the Laplace Transform of the waveform $v(t)$. [03]

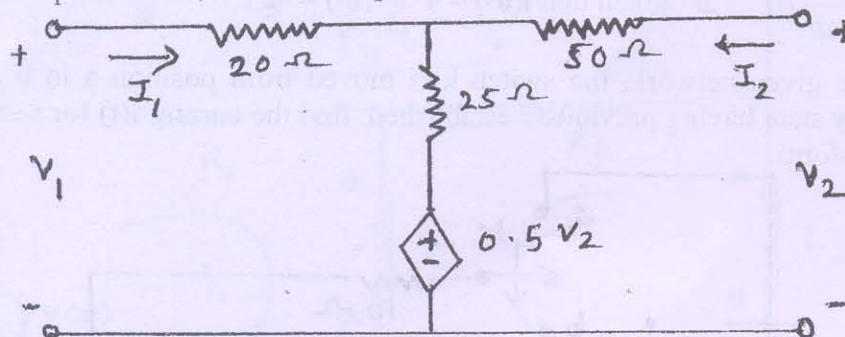


SECTION - II

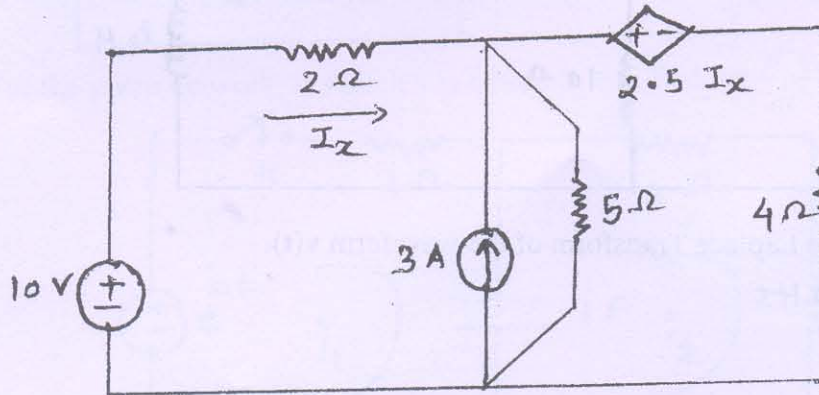
Q-4 (a) For the network shown in figure, Determine the current i_1 using Source Transformation Method. [07]



Q-5 (a) Find the Z parameters, for the Two port network shown in figure. [07]

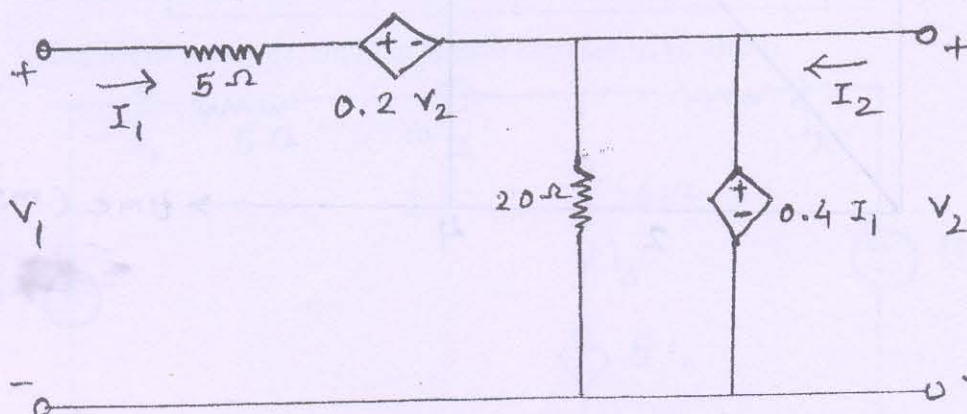


(b) Find the current I_x in the circuit shown in figure using Superposition theorem. [07]



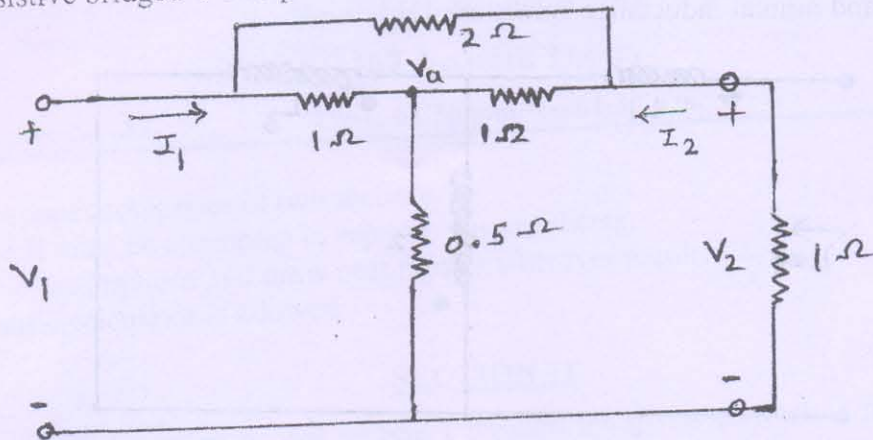
OR

Q-5 (a) Find the $ABCD$ parameters for the Two port network shown in figure. [07]



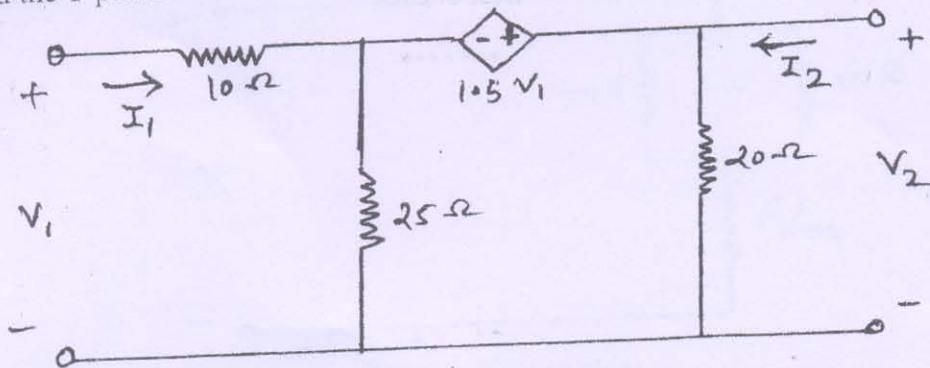
(b) A resistive bridged-T two port network shown in figure, find the G_{12} and α .

[07]



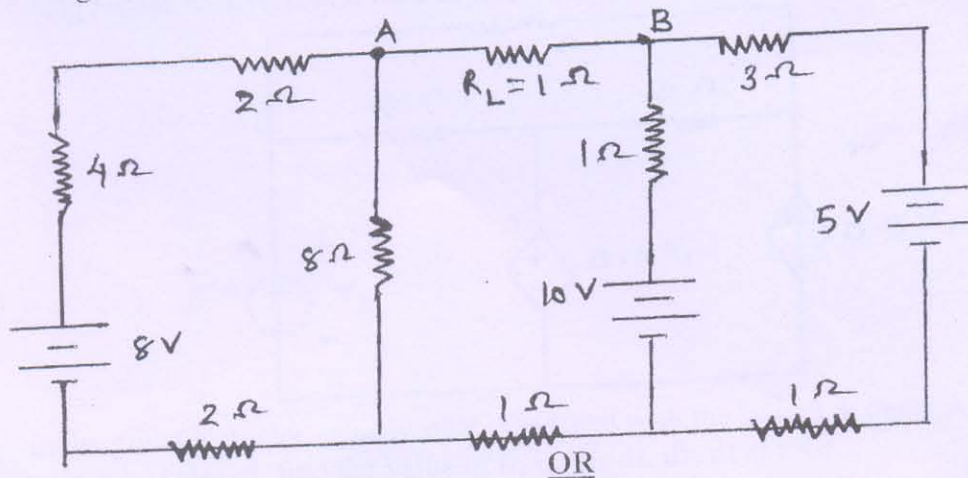
Q - 6 (a) Find the Y parameters for the Two port network shown in figure.

[07]



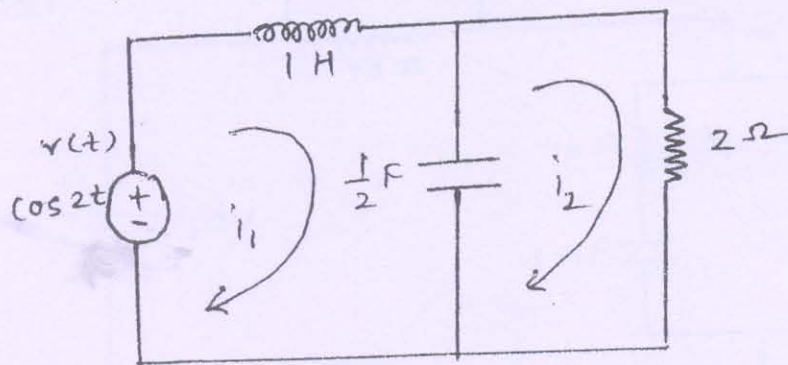
(b) In the given network, find the current through R_L using Thevenin's theorem.

[07]

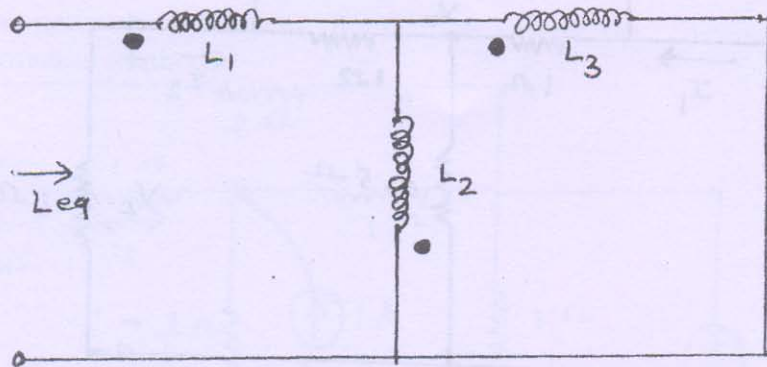


Q - 6 (a) Find the steady state current $i_2(t)$ for the network shown in figure.

[07]



- (b) Find the equivalent inductance of the given network, where all self inductance values are 2H and mutual inductance values are 1H . [07]



Best of Luck
